

1 Capture schedule

Benign traffic on 6-8 February, every attack on 10-12 February. Each class occupies one contiguous window of the capture.

40 h with no labelled traffic between the two blocks

Section 3.3

2 Shortcut

Position in the capture becomes a near-perfect class predictor, and eight behavioural columns carry the same information.

StartTime alone: 0.9970 accuracy, against 0.1942 for the majority class

Section 6.2

3 Random split

The standard protocol hides it. The nine-class task is solved and the leaderboard carries almost no information.

LightGBM at macro-F1 1.0000 and XGBoost at 0.9999; 11 of 45 pairs not separable

Section 6.1

the standard remedy, and on this corpus it does not remove it

4 Temporal split

The usual remedy, applied within each class. It does not remove the shortcut: a class sits in one window, so ordering inside that window cannot break the association.

the same probe still scores 0.9862

Section 6.2

5 Audit

Exclude what does not survive the change of protocol, transferability below 0.50 among columns that are individually predictive, and what identifies a capture file.

13 of 67 columns excluded; permutation importance below $1e-4$ for the eight the ratio rule finds

Section 4.3

6 Audited benchmark

Removing them costs almost nothing under a random split, which is what shows the shortcuts were redundant. The temporal ranking is what moves.

largest stratified loss 0.0018; both boosted ensembles leave the top

Section 6.3

 the shortcut is present

 the shortcut is removed, by feature exclusion rather than by the protocol